Aircraft Maintenance Prediction

Section G2 Group 10

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Business Problem

<u>Problem Statement</u>:
A good maintenance plan is one of the key drivers to ensuring an aircraft is fit for function.
Maintenance can be largely categorized into two:

- *Planned maintenance*
- *Unplanned maintenance*

Predictive maintenance can allow for accurate lifetime prediction of components therefore pre-emptively replacing components that will be damaged in between maintenance cycles potentially shifting the maintenance time spent in the unplanned region, over to the planned region.

	Without ML	**With ML**
Core Idea	Scheduled inspections and manual fault checks	Predict maintenance risks using historical flight telemetry
Pattern Detection	Limited to fixed rules and thresholds	Learns complex multivariate time-series patterns
Data Used	Manual inspections and safety checks	Per-second aircraft sensor and maintenance data
Adaptability	Rules updated manually	Adapts to changing flight behaviour automatically
Accuracy	May miss early warning signs	Improves anomaly and maintenance prediction
Scalability	Difficult across large fleets	Scales across thousands of flights
Output	Reactive maintenance actions	Predictive maintenance alerts and early warnings

Predictive maintenance model enables the airline to proactively address fleet maintenance needs, reduced inventory cost, prevent potential flight delays and optimise engineering productivity

Methodology

Repeat if necessary

EDA and Data cleaning

- Exploration of dataset
- Data cleaning and aggregation
- Missing value analysis
- Class imbalance check
- Correlation matrix to identify redundant features

Feature Engineering and Pre-processing

- Feature engineering such as rolling stats (mean, std, max), interaction features
- Feature selection and normalization (MinMax or StandardScalar)
- Training, validation, test, unseen data split

Model Building and Selection

- Exploration of different models with minimal tuning to review effectiveness
- Selection of final model

Tuning and Refinement

- Hyperparameter tuning using Optuna
- Ensemble / Stacking models

Evaluation of Model and Conclusion

- Evaluation of Recall, AUC, F1 Score, etc.
- Conclusion and discussions

Train / 70%	Validation / 15%	Test / 15%

Goal is to predict maintenance required based on a fixed set of labels, the ML approach will be that of a classification problem. By analyzing flight sensor patterns before and after maintenance events, the model will predict maintenance-related risk states, helping airlines enable earlier inspections and proactive maintenance planning.

Dataset and Proposed ML Solution

This dataset uses flight records and unplanned maintenance logs from a Cessna 172 aircraft, sourced from the National General Aviation Flight Information Database (NGAFID)

29k *flights* X **4,400** *Median timesteps per flight* = **Multivariate time series** *of sensor/flight ⇒ 113 million records*

36 *Unique maintenance label type*

23 *Sensor data + 7 Metadata columns*

Sensor Data

Column Name	Description
volt1	Main electrical system bus voltage (alternators and main battery)
volt2	Essential bus (standby battery) bus voltage
amp1	Ammeter on the main battery (+ charging, - discharging)
amp2	Ammeter on the standby battery (+ charging, - discharging)
FQyL	Fuel quantity left
FQyR	Fuel quantity right
E1FFlow	Engine fuel flow rate
E1 OilT	Engine oil temperature
E1 OilP	Engine oil pressure
E1 RPM	Engine rotations per minute
E1 CHT1	1st cylinder head temperature
E1 CHT2	2nd cylinder head temperature
E1 CHT3	3rd cylinder head temperature
E1 CHT4	4th cylinder head temperature
E1 EGT1	1st Exhaust gas temperature
E1 EGT2	2nd Exhaust gas temperature
E1 EGT3	3rd Exhaust gas temperature
E1 EGT4	4th Exhaust gas temperature
OAT	Outside air temperature
IAS	Indicated air speed
VSpd	Vertical speed
NormAc	Normal acceleration
AltMSL	Altitude miles above sea level

Input Features (X)

- 23-channel multivariate time series (engine, fuel, electrical, flight performance)

Binary Classification Problem

Flight Metadata

Column Name	Description
Master Index	Primary Key to link to the flight records
Before After	Whether or not a flight occurred before or after maintenance. Some flights occurred during maintenance (generally test flights)
Date Diff	Days before or after the maintenance period
Flight Length	Number of flight recordings for the whole flight (duration of flight in seconds)
Label	Maintenance Issue
Hierarchy	Maintenance Issue Hierarchy, if NaN then Other.
Number of Flights Before	The Nth flight before maintenance, with 0 representing the flight immediately before and -1 representing not applicable.

Target Output (Y)

- Target: P(RUL > t days), binary, from before_after + date_diff

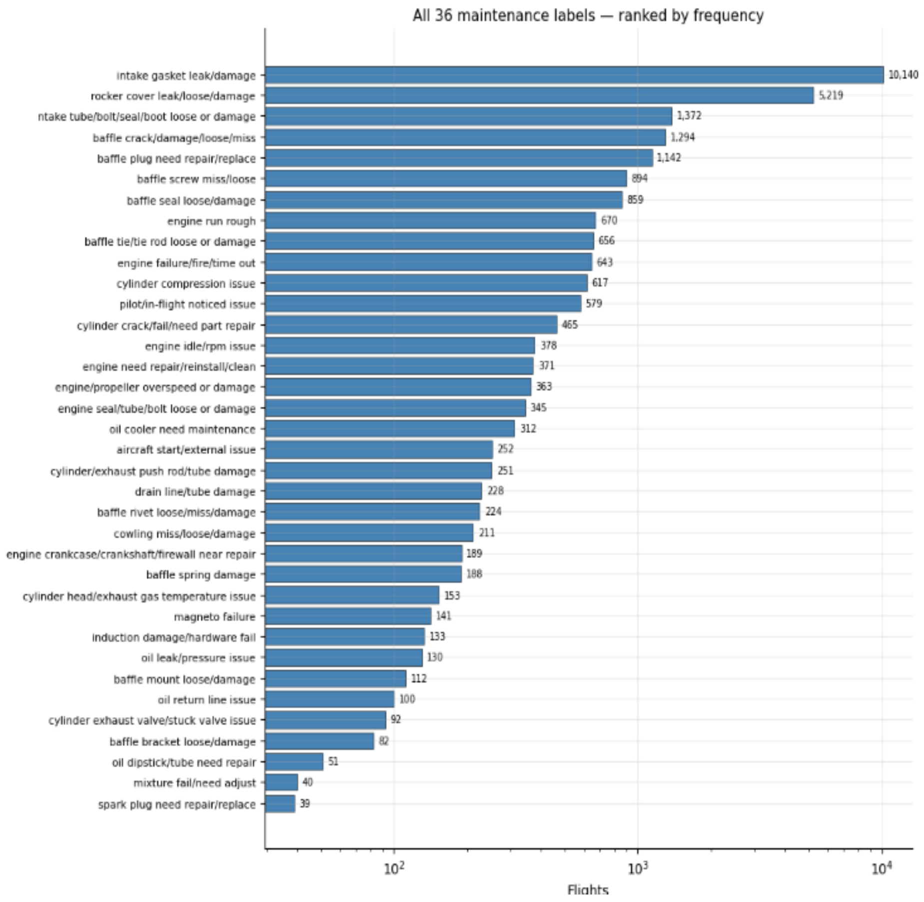
RUL = Remaining Useful Life

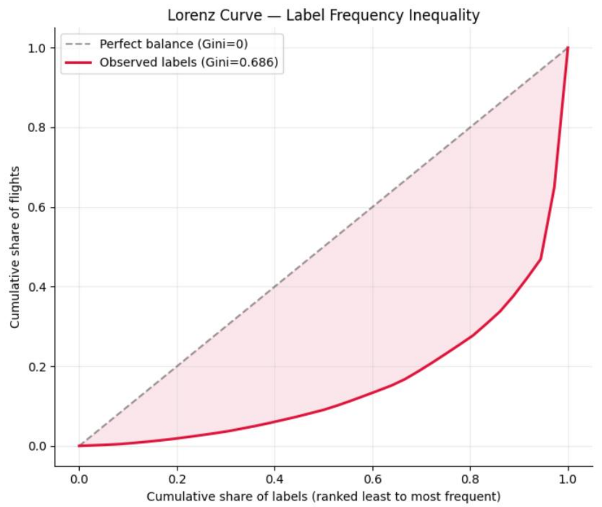
[Link: Aviation Maintenance Dataset from the NGAFID]()

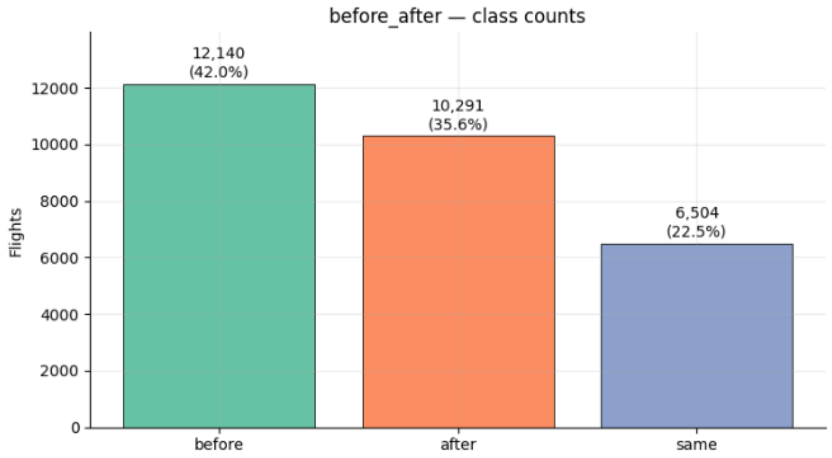
EDA Summary

```
=================================================================
  NGAFID DATASET — KEY METRICS
=================================================================
  Total flights             : 28,935
  Sensor channels           : 23
  Unique maintenance types  : 36
  Avg flight length         : 3,879 timesteps
  Flight length range       : 12 – 30059 timesteps

Highly-correlated channel pairs  (|r| > 0.9):  25
```

```
hierarchy null rate: 63.43%
Distinct hierarchy values: 4

Hierarchy distribution (incl. NaN):
hierarchy
NaN         18354
baffle       5451
engine       2959
cylinder     1578
oil           593
```

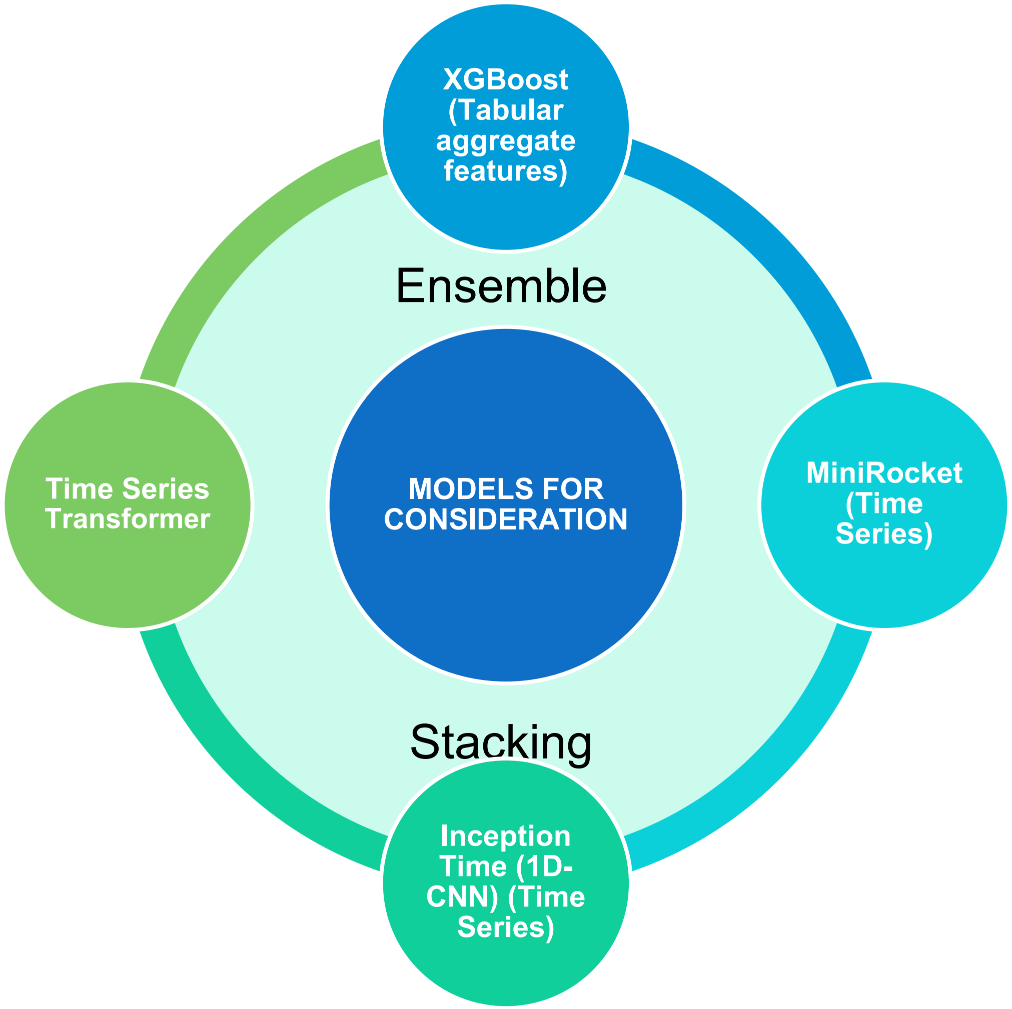






- Dataset spans 28,935 flights across 36 distinct maintenance event types, concentrated 35% on intake gasket leaks alone, showing maintenance burden is unevenly distributed
- Before/ After/ Same label distribution is 42% / 36% / 22% → mild imbalance (class ratio 1.87x)
- Severe long tail in 36-class label (Gini 0.686) → > 0.6 is severe, > 0.7 is extreme long-tail
- 63.4% of rows have no hierarchy value. Hierarchical multi-task learning is not feasible with this label
- Correlation analysis suggests sensor redundancy: 23 channels → 11 after deduplication

ML Models and Performance Evaluation



Evaluation Criteria

Recall (Sensitivity)

A false negative clears an at-risk aircraft to fly again, so the failure risk falls on the next sortie. High recall minimizes this safety-critical miss.

Precision

Avoids too many unnecessary maintenance calls (false positives), which are costly and reduce fleet availability.

F1-Score

Harmonic mean of Precision and Recall. Penalizes models that sacrifice one for the other, balancing safety against operational disruption.

ROC-AUC / PR-AUC

Threshold-independent measures for comparing all models fairly. PR-AUC supplements ROC-AUC when class distribution is imbalanced.

Confusion Matrix

Full breakdown of TP, TN, FP, FN mapping directly to operational outcomes: correctly flagged, correctly cleared, unnecessary inspections, and missed faults.

Inference Time

Time to score a single completed flight at deployment. Must be fast enough to inform the pre-flight check before the aircraft's next sortie.

Analysis and Model Building (Potential Limitations)

Potential Limitations	Description	Addressed By
Limited to Only Cessna 172 plane type	The dataset contains sensor and flight data of only Cessna 172 plane type	As this study is meant as a proof of concept and preliminary study for predictive maintenance in the aviation industry, it is not meant to be comprehensive. In order to extend to other plane types, a different ML model will need to be built for every plane since they will have different sensors characteristics.
The dataset is only for US	The dataset is only for US Cessna 172 planes. There may be deviations in environmental factors e.g. temperature, humidity etc., which may be outside the range of the training dataset	The models built for this study will require field validation for every new environment it will operate in. This will in turn provide insights on whether the features and dataset are sufficient to successfully implement a predictive maintenance plan or if extra sensor will be required.
Information is loss due to grouping	Due to grouping, the breakdown of the required maintenance is lost.	The purpose of the study is not to do away with pre-flight inspections, but to allow for better maintenance planning via targeting of the selected group.
Class Imbalance in Label	2 out of 36 labels contain fewer than 50 samples. Some classes having thousands of flights and others having less than one hundred.	Rare labels will be grouped into broader categories, improving sample representation per class and focal loss to be applied during training to place greater emphasis on under-represented classes and reduce bias toward majority classes.

Application and Future Scope

- The study will be conducted to evaluate the feasibility of predictive maintenance for aircraft targeting audience such as airline operators, aviation safety regulators and OEMs.
- The study, while limited in scope due to data from a singular plane type, allows for a controlled scale study from which to carry our further validation studies.
- The next step is to observe over a period of time whether a new maintenance strategy based on the model will be able to successfully reduce unplanned downtime risk and save cost for the operator.
- As this study is conducted only for a small population of private jets, the strategy can be modified such that a “just-in-time” model can be established. This will serve as a proof of concept, beyond which the same study can be extended to commercial plane types such as Airbus A320, Boeing 737 Max etc.